

The Impossibility of a Paretian Liberal

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Introduction

Last time we looked at ways out of Arrow's impossibility theorem.

The impossibility theorem said that some intuitively plausible principles are in fact inconsistent.

So that suggests we should look at how plausible those principles actually are.

Today we're going to question one particularly plausible looking principle: Pareto Optimality.

In 1970, Amartya Sen asked, can we have a social choice function that respects:

- Pareto optimality (P)
- Individual liberty (L)
- Unrestricted domain (U)

His answer: **No**.

The Pareto principle seems like a minimal requirement for respecting individual preferences:

- If everyone prefers A to B, society should prefer A to B.

But Sen shows this can conflict with another attractive principle:

- Individuals should have decisive control over some personal choices.

The Pareto Condition in Arrow

Condition P: If every individual prefers x to y , then society must prefer x to y .

This was one of Arrow's four conditions, and it seemed the least controversial. Well, perhaps almost the least after non-dictatorship.

- After all, if literally everyone thinks A is better than B , shouldn't society agree?

In the context of Arrow's theorem, P seemed like it captured something minimal about the idea that the group ranking was really the ranking **of the group**.

If everyone prefers x to y , in what sense is a ranking that prefers y to x that group's ranking?

The Pareto principle also looks like a good bit of individualism.

We don't let some abstract construct, like the social will, override what literally every individual wants.

Surprisingly, Sen shows that it also conflicts with other individualistic principles.

Sen's Liberal Condition

Condition L (Liberalism): For each individual i , there is at least one pair of alternatives (x, y) such that:

- If i prefers x to y , then society should prefer x to y
- If i prefers y to x , then society should prefer y to x

In other words: Each person should be decisive over at least one pair of alternatives.

The pairs should be things that are genuinely **personal** choices, such as:

- Whether to paint your walls pink or white;
- Whether to sleep on your back or stomach; or
- Whether to read a particular book.

These are choices where, plausibly, your preference should be decisive regardless of what others think.

Condition L is **extremely weak**.

Compare the principle that says I get to choose whether I have salad or burrito for lunch.

That implies that whatever everyone else does, I'm decisive over that. So I'm decisive over any number of pairs of full social outcomes.

Sen's condition just says I'm decisive over one pair. It's like saying there is one possible set of choices everyone else makes, and if they make them, I get to decide whether burrito or salad. This looks like a really weak claim.

Sen also considers an even weaker condition:

Condition L* (Minimal Liberalism): There are at least two individuals such that for each of them there is at least one pair of alternatives over which they are decisive.

Even this minimal condition—giving just two people control over one personal choice each **in one situation**—leads to impossibility.

Sen's Impossibility Theorem

Theorem: There is no social decision function that can simultaneously satisfy:

- Unrestricted Domain (U)
- Weak Pareto Principle (P)
- Minimal Liberalism (L^*)

This is surprising: two principles that both seem to respect individual preferences turn out to be incompatible.

Sen does not use some of the assumptions that seemed potentially controversial in Arrow.

- He does not assume that we end up ranking all the options; just that there is one option that no option is better than.
- He does not assume **Independence of Irrelevant Alternatives** (IIA), or anything like it.

But he does assume this Minimal Liberalism, which Arrow did not.

Lady Chatterley's Lover

There are two individuals and three alternatives:

- Individual 1 is a **prude**; and we'll call them Prude.
- Individual 2 is not; and we'll call them Lewd.

There is one copy of D.H. Lawrence's *Lady Chatterley's Lover*.

The Three Alternatives

- **x**: Individual 1 reads the book
- **y**: Individual 2 reads the book
- **z**: No one reads the book

Question: What should society choose?

Individual 1 is a prude. He thinks the book is corrupting.

His preference ordering is: $z > x > y$

- Most preferred: No one reads it (z)
- Second: He reads it himself (x)
- Least preferred: Individual 2 reads it (y)

Why does he prefer x to y? “Prudes, I am told, tend to prefer to be censors rather than being censored.”

Individual 2 wants someone to read the book.

His preference ordering is: $x > y > z$

- Most preferred: Individual 1 reads it (x)
- Second: He reads it himself (y)
- Least preferred: No one reads it (z)

Why does he most prefer x? He “takes delight in the thought that prudish Mr. 1 may have to read Lawrence.”

If they both want to read the book, that's a clash of interests and we don't assume liberalism can help.

But if it's fixed that the other person is not reading the book, so the only choice is that individual reads or no one does, the individual is decisive.

Consider the choice between **x** and **z** (Person 1, Prude, reading it vs. no one reading it):

- This is a personal choice for Prude.
- He prefers **z** (no one reads) to **x** (he reads).
- By Condition L^* , society should respect his preference.
- Therefore: Society should prefer **z to x**.

Consider the choice between **y** and **z** (Person 2 (Lewd) reading it vs. no one reading it):

- This is a personal choice for Lewd.
- He prefers **y** (he reads it) to **z** (no one does).
- By Condition L^* , society should respect his preference.
- Therefore: Society should prefer **y to z**.

Now consider the choice between **x** and **y**:

- Individual 1 prefers **x** to **y** (he ranks $x > y$).
- Individual 2 prefers **x** to **y** (he ranks $x > y$).
- So, **everyone** prefers **x** to **y**.
- By the Pareto Principle, society must prefer **x** to **y**.

We have derived:

1. Society prefers z to x (from liberalism for Individual 1).
2. Society prefers y to z (from liberalism for Individual 2).
3. Society prefers x to y (from Pareto).

This gives us: $y > z > x > y$.

So there is no option such that no option is better than it, so society can't make a choice.

Understanding the Result

The problem arises because individuals have preferences about what **others** should do:

- Individual 1 cares whether Individual 2 reads the book.
- Individual 2 cares whether Individual 1 reads the book.

These are called **externalities** or **nosy preferences**.

The intuition behind liberalism is:

- Your personal choices should be yours alone.
- Others shouldn't be able to override what you do in your private sphere.

But if others have strong preferences about your choices, respecting everyone's preferences (Pareto) conflicts with protecting your personal sphere (Liberalism).

You might think: “This only happens with weird preference configurations.”

But Sen argues these situations are not uncommon:

- Moral busybodies who care what others read, watch, or do.
- People with interdependent preferences (keeping up with the Joneses).
- Cases where others' choices affect your welfare (smoking, loud music).

The last case relates to the Prisoners' Dilemma, which we'll come back to later in the course, and where we also see a clash between individual interests and group interests, as long as externalities exist.

Responses to Sen

One major line of criticism (Nozick, Sugden, others): Sen has misunderstood what liberty requires.

The objection: Liberty isn't about which outcomes emerge—it's about who controls the decision procedure.

Individual 1 should have **control** over whether he reads the book.

Individual 2 should have **control** over whether she reads it.

If each has control, that's all liberty requires—regardless of what outcome emerges.

Robert Sugden (1981) puts it forcefully:

Mill would have agreed that “there are certain personal matters in which each person should be free to decide what should happen”; but would he have agreed that “in choices over these things whatever he or she thinks is better must be taken to be better for society as a whole”?

The first is about **procedures**. The second is about **end-states**.

Why should a liberal have to claim procedurally-protected outcomes are always best?

(The discussion here follows Sen's 1983 paper “Liberty and Social Choice”, and the references are to books and papers Sen discusses in it.)

Sen's minimal liberty condition (ML) doesn't actually require that individuals themselves make the choice.

It only requires that what emerges **matches what they would have chosen**.

This is compatible with both:

- **Direct liberty:** The individual actually exercises control; and
- **Indirect liberty:** Someone else chooses on the basis of what the individual would choose.

Example: Unconscious Ed and the doctor choosing treatment B because that's what Ed would choose.

Modern society cannot be organized so each person directly controls all decisions affecting their “personal sphere.”

But we still make judgments like:

- “The police preventing mugging serves my liberty” (I would choose not to be mugged); or
- “Traffic laws serve my liberty” (I would choose not to be hit by wrong-way drivers).

These are liberty considerations, not just welfare considerations.

To see liberty **only** as direct control leaves out too much.

The Trading Solution

Perhaps the most common response: Let individuals **trade** their rights away.

In Lady Chatterley's Lover:

- Prude and Lewd make a contract. It says that Prude agrees to read it if Lewd doesn't read it.
- Both prefer x (Prude reads) to the "liberal" outcomes.
- This gets us to the Pareto optimal outcome.

Problem solved?

Sen identifies several distinct issues with the trading solution. We'll focus on three.

1. **Legitimacy:** Should rights be tradeable in this way?
2. **Contract-means:** Will people accept contract enforcement?
3. **Instability:** Can the contracted outcome be sustained?

Let's examine each.

1. The Legitimacy Issue

Is it legitimate to make contracts that alienate your freedom in your personal sphere?

Mill argued: “It is not freedom to be allowed to alienate his freedom.”

- But that was about slavery.
- Contracts about what to read seem different.

1. The Legitimacy Issue

More concerning: Should such contracts be **publicly enforceable**?

The enforcer checking whether Prude reads *Lady Chatterley's Lover* each morning is “morally problematic, aside from being deeply chilling.”

Trading away privacy rights is getting closer to things we think shouldn't be tradable.

2. The Contract-Means Issue

Even if you want the outcome, do you want it **through a binding contract**?

In relationships: Getting someone to behave well through a contract may defeat the purpose.

“I want you to be faithful because you want to be, not because you signed a contract.”

The **means** of achieving an outcome affects its value.

Still, I'm not sure it's a problem here. Will Lewd really be upset if Prude reads because he's contractually obligated?

3. The Instability Issue

This is the most fundamental problem.

Even if:

1. The contract is legitimate;
2. Both parties want the outcome; and
3. Both parties accept that the contract is valid:

They still both have an incentive to break the contract.

Remember the cycle in Lady Chatterley:

- Liberalism says: $z > x$ (respect Prude's choice)
- Liberalism says: $y > z$ (respect Lewd's choice)
- Pareto says: $x > y$ (unanimous preference)

The contract might get us to x , but:

- Given x , Prude wants to move to z (his liberal right).
- Given z , the book is lying around unread, Lewd wants to move to y (his liberal right).
- And now we're back in the Pareto-suboptimal outcome.

We're still in a cycle, and there may be no stable outcome at all.

The possibility of Pareto-improving contracts:

- Raises new legitimacy questions;
- Leads to instability; and
- Leaves the fundamental conflict unresolved.

Is he right? I'll leave that for you to think about!

Implications

Sen's impossibility shows that two principles both claiming to respect individual preferences can conflict:

- **Pareto:** Respect unanimous preferences;
- **Liberty:** Respect individual preferences over personal spheres.

This conflict is fundamental, not a technical glitch.

Central issue: What role should preferences play in moral and political philosophy?

Traditional view: If everyone prefers x to y , x must be better.

Sen's result: This conflicts with respecting individual freedom.

Maybe some preferences (nosy, envious, meddlesome) shouldn't "count"—even if they're genuinely held.

One theory of well being, popular among economists, is that making people better off just is giving them more of what they prefer.

This seems wrong for moral preferences (like the one Sen discusses at the start of *Liberty and Social Choice*) or other-directed preferences (like Lewd's preference that Prude reads the book).

One way to think about Sen's point is that the same preferences that raise problems for this theory of welfare also raise problems for the idea that Pareto-optimality principle is a way of respecting individual views.

Sen's impossibility doesn't tell us how to resolve the conflict.

But it clarifies what we're dealing with:

- We can't have unrestricted domain + Pareto + even minimal liberty.
- So we must give up one of these.
- In some sense we must not respect some preferences.
- Either Prude must read the book he does not want to read (violating his self-directed preferences), or we end up in a Pareto-suboptimal outcome (violating both party's other-directed preferences).

It's a hard policy decision to balance these.

We'll move on from aggregating preferences to aggregating beliefs, which leads to its own set of problems.